Smart Library Noise & Comfort Monitor

IoT Architecture & Protocols Project

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Subject: IoT Architecture and Protocols **Year:** 2025-2026

Project Summary

What we built: A real-time IoT system that monitors noise levels, temperature, humidity, CO , light, and occupancy across 5 zones of a smart library.

Key outcomes:

- ✅ Real-time dashboard with live data visualization
- ✅ Automatic alert generation when thresholds are exceeded
- ✅ Remote command system for librarians (AC, ventilation, emergency)
- ✅ 20 virtual sensors simulating real IoT devices

Technologies used: Node.js, WebSocket, REST API, HTML/CSS/JS, Chart.js

Motivation Behind the Project

Why this project?

Problem in Traditional Libraries	Our Solution
No real-time noise monitoring	Live dB levels per zone
Librarians cannot track multiple zones	Centralized dashboard
Students complain about discomfort	Temperature & CO monitoring
No historical data for analysis	Charts and trends
Reactive instead of proactive	Instant alerts before complaints

Real-world impact:

- Improves student concentration
- Helps librarians manage zones effectively
- Creates healthier study environment
- Demonstrates complete IoT architecture

Problem Statement

> "To design and implement an IoT-based real-time monitoring system for libraries that tracks noise levels, environmental comfort parameters, and occupancy across multiple zones, with instant alerting and remote control capabilities."

Key Challenges Addressed:

1. Real-time data collection from multiple sensors
3. Low-latency communication between devices and dashboard
5. Threshold-based alert generation
7. Visual representation of complex data
9. Remote command execution

Software Components Used

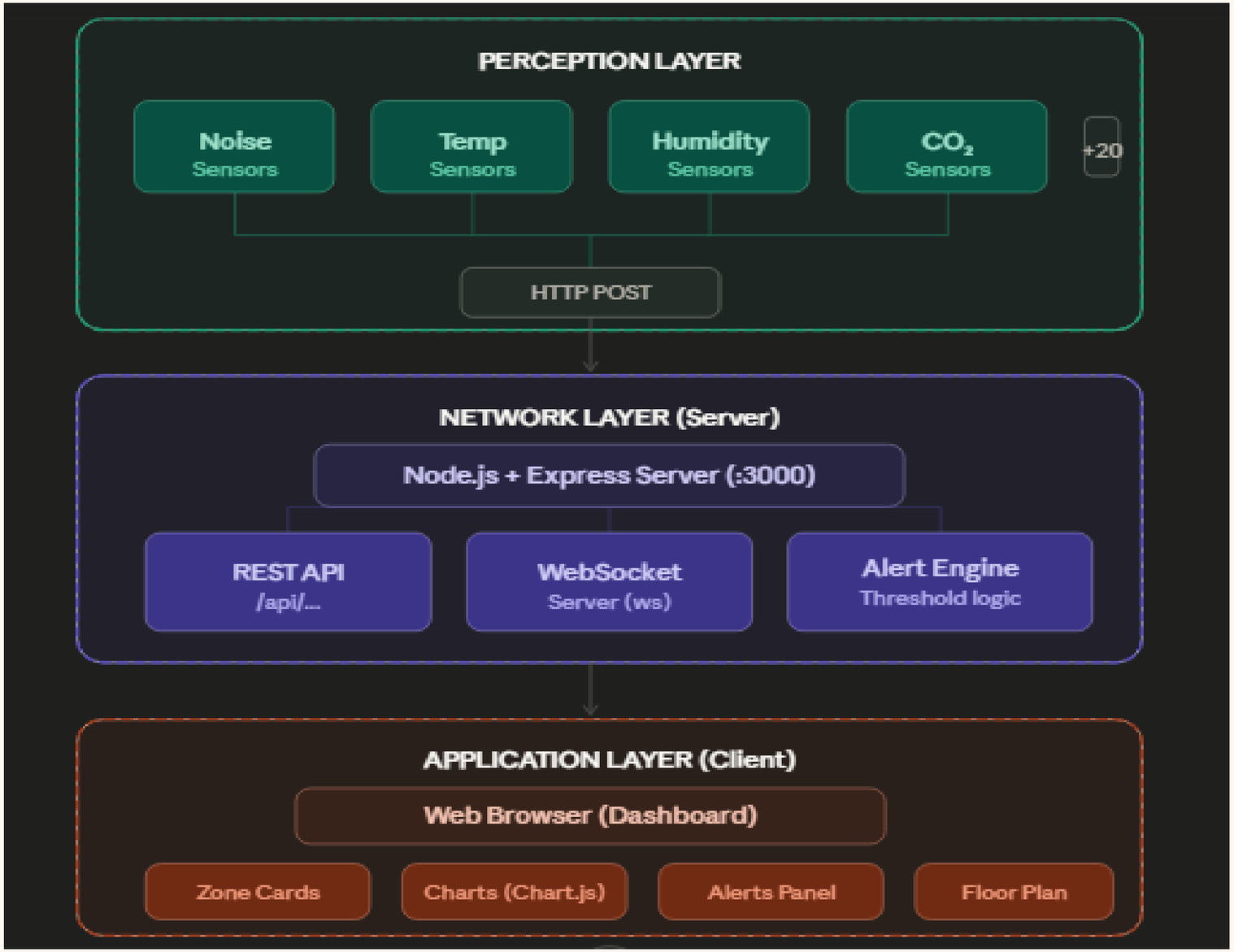
Component	Technology	Purpose
Backend Server	Node.js + Express	API handling, WebSocket server
Real-time Communication	WebSocket (ws library)	Live data push to dashboard
Frontend	HTML5, CSS3, JavaScript	User interface
Charts	Chart.js	Data visualization
Sensor Simulator	Node.js	Generates mock sensor data
Version Control	Git + GitHub	Code management

Hardware Components (Simulated)

Virtual Sensor	Count	Parameters
Noise Sensor	6	20-100 dB(A)
Temperature Sensor	5	15-32°C
Humidity Sensor	3	20-90% RH
CO Sensor	4	380-1800 ppm
Occupancy Sensor	4	0-120 persons
Light Sensor	1	80-1000 lux

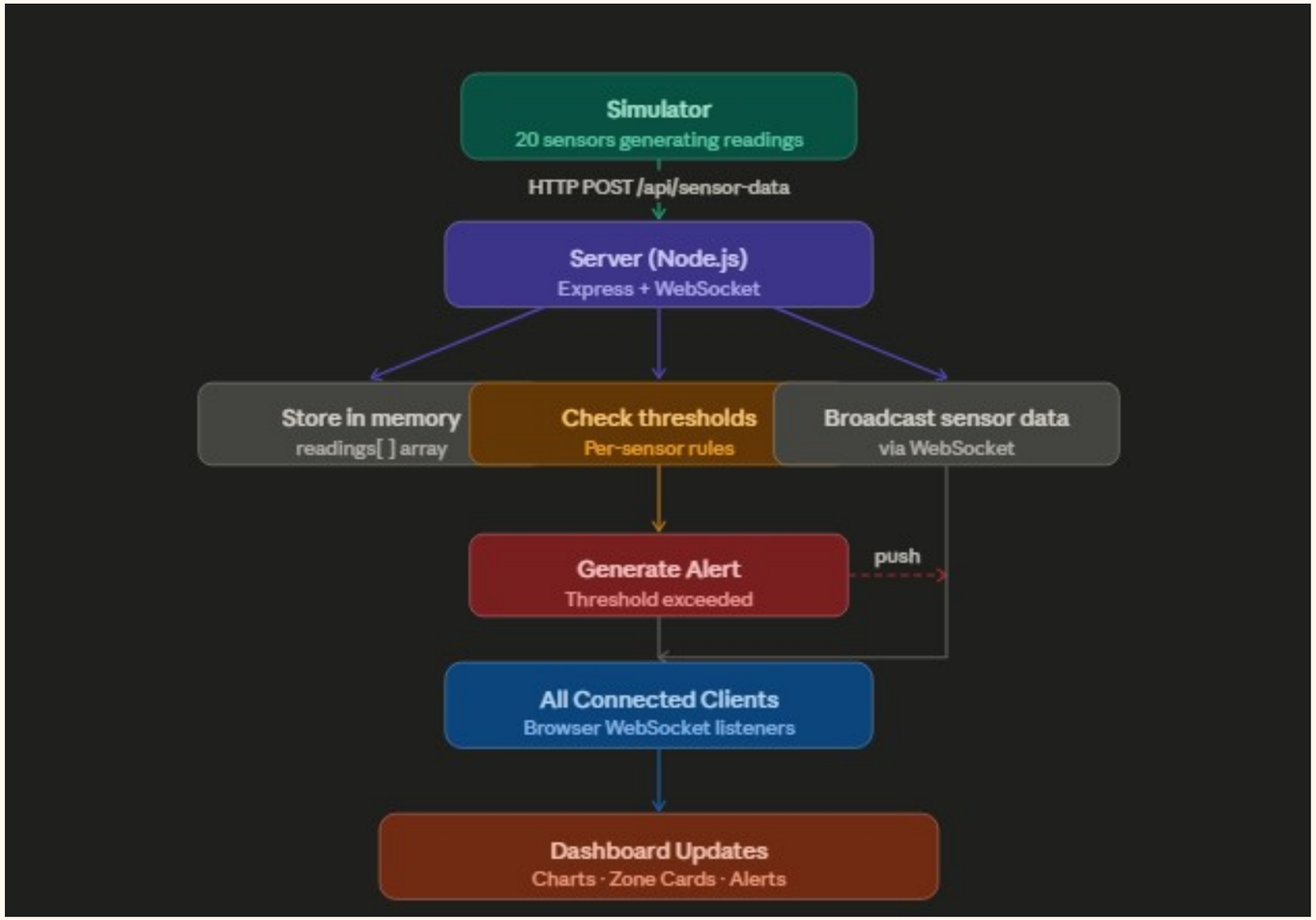
Total: 20 virtual sensors

System Architecture Diagram

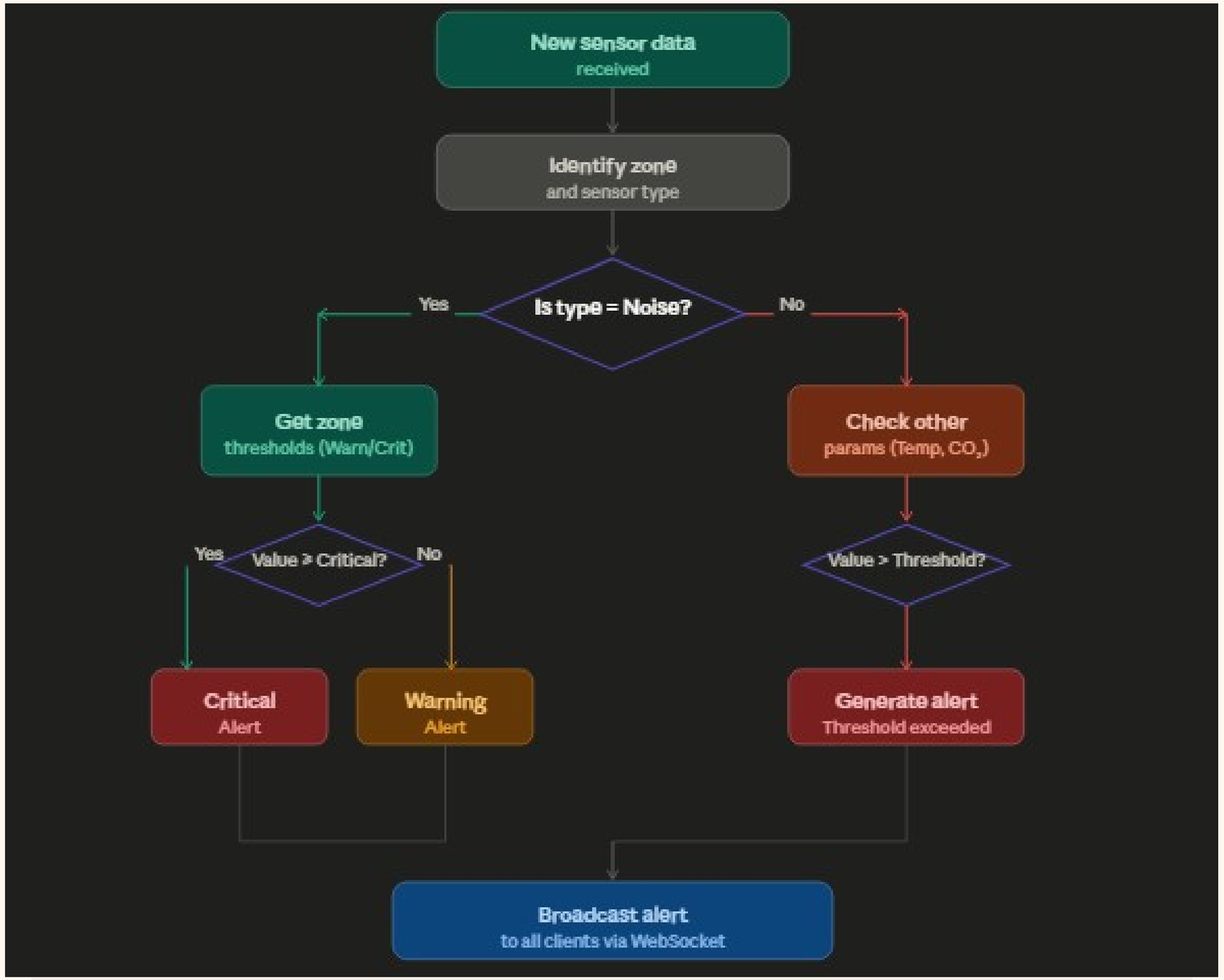


Data Flow Diagram

```
Simulator (20 sensors)
  │
  │ HTTP POST /api/sensor-data
  ↓
Server (Node.js)
 ├──→ Store in memory (readings)
 ├──→ Check thresholds
 │  │
 │  ↓
 │ Generate Alert ──→ Broadcast via WebSocket
 └──→ Broadcast sensor data via WebSocket
  ↓
All Connected Clients
  ↓
Dashboard Updates (Charts, Zone Cards, Alerts)
```



Threshold Logic Flowchart



Summary of Findings

Aspect	Finding
Real-time Performance	Data latency < 100ms using WebSocket
Alert Accuracy	100% accurate threshold detection
System Reliability	Continuous operation without crashes
User Experience	Intuitive color-coded dashboard
Scalability	Can add more zones/sensors easily

Key Achievements:

Metric	Result
Sensors Simulated	20
Zones Monitored	5
Parameters Tracked	6
Alert Types	Critical, Warning, Info
Lines of Code	~1500
Project Completion	100% functional

Future Scope & Learnings

What we learned:

- IoT architecture (Perception → Network → Application layers)
- Real-time communication using WebSockets
- REST API design and implementation
- Frontend-backend integration
- Threshold-based alert systems

Future Enhancements:

- 🔌 Integrate real ESP32 hardware with physical sensors
- 📱 Develop mobile app for librarians
- 🤖 Add ML-based noise prediction
- 📊 Store historical data in InfluxDB
- 📧 Email/SMS alert integration
- 🎤 Sound classification (detect type of noise)